

Variational Autoencoders

Experimental Report

Detailed documentation of implementation steps, experimental results, and conclusions for VAE-based generative models.

Part of a comprehensive comparison with Denoising Diffusion Probabilistic Models (DDPMs).

May 24, 2026

Report Structure

- **Part I — Background:** Theory of VAEs and model architecture
- **Part II — Experimental Setup:** Fixed hyperparameters and dataset
- **Part III — Experiments:** Per-experiment results, analysis, and figures (losses, samples, reconstructions, latent spaces)
- **Part IV — Conclusions:** Summary of findings and key differences

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1 Introduction

This report documents a systematic experimental study of **Variational Autoencoders (VAEs)** applied to image generation. Each experiment targets a specific design choice — conditioning strategy, attention mechanism, perceptual loss, or capacity — and evaluates its effect on generation quality, reconstruction fidelity, and latent-space structure.

The document is designed to serve as one half of a broader **comparative analysis between VAEs and Denoising Diffusion Probabilistic Models (DDPMs)**, enabling an apples-to-apples comparison of two dominant paradigms in deep generative modelling.

1.1 Purpose of the Comparison

VAEs and DDPMs represent fundamentally different inductive biases:

- **VAEs** learn an explicit, amortised posterior $q_\phi(\mathbf{z}|\mathbf{x})$ and optimise a tractable Evidence Lower Bound (ELBO), trading reconstruction sharpness for a structured, continuous latent space.
- **DDPMs** learn to reverse a fixed Markov diffusion process, achieving high sample quality at the cost of slow, iterative inference.

Both model families are trained on the **same architectural backbone**, the same dataset, and the same fixed hyperparameters, so that differences in metrics reflect the generative paradigm rather than implementation disparity.

2 Background: Variational Autoencoders

Content placeholder

This section will contain:

- A concise derivation of the ELBO objective
- The reparameterisation trick
- KL-divergence regularisation and its role
- Perceptual / feature-matching losses
- Conditional generation strategies (class embeddings, AdaGN, concatenation)

2.1 Model Architecture

Content placeholder

This section will contain:

- Encoder–decoder design (U-Net style with residual blocks)
- Channel multiplier schedule
- Attention insertion points
- Conditioning injection: AdaGN vs. channel-concatenation
- Comparison diagram vs. DDPM denoiser architecture

3 Experimental Setup

3.1 Dataset

All experiments use a 10-class image dataset with resolution 32×32 pixels.

3.2 Fixed Hyperparameters

The following hyperparameters are held constant across *all* experiments to ensure fair comparison:

Fixed Hyperparameters (all experiments)

```
{
  "image_size"      : 32,          # Input / output resolution (
    px)
  "num_classes"     : 10,          # Number of conditioning
    classes
  "latent_dim"      : 128,         # Dimensionality of latent
    space z
  "base_channels"   : 64,          # Feature-map width at first
    encoder level
  "channel_mults"   : [1, 2, 4],   # Channel multipliers per
    resolution level
  "num_res_blocks"  : 2,           # Residual blocks per
    resolution level
  "batch_size"      : 32,          # Training batch size
  "lr"              : 0.0002,     # Adam learning rate
}
```

3.3 Model Variants

- **VAE** — Vanilla VAE with reconstruction + KL loss only.
- **PercepVAE** — VAE augmented with a perceptual (feature-matching) loss.
- **cVAE-AdaGN** — Conditional VAE using Adaptive Group Normalisation to inject class information.
- **cVAE-Concat** — Conditional VAE using channel concatenation of the class embedding.

3.4 Evaluation Metrics

FID Fréchet Inception Distance — measures the distributional gap between real and generated images in Inception feature space. *Lower is better.*

IS (Inception Score) Evaluates both the quality and diversity of generated images. Reported as mean $\pm$ standard deviation over multiple splits. *Higher is better.*

4 Experiment 1

4.1 Motivation

The first experiment establishes the **baseline** performance of four VAE variants that differ in their conditioning and loss strategy, without self-attention. This provides a reference point for subsequent ablations.

4.2 Hyperparameters

Experiment 1 — Variable Hyperparameters

```
{
  "epochs"      : 50,
  "kl_weight"    : 0.001,    # Weight on KL-divergence term
  "percep_weight": 0.1,      # Weight on perceptual loss
  "use_attention": false,    # Self-attention disabled
  "attn_heads"   : 1,
}
```

4.3 Evaluation Results

Table 1: Experiment 1 — Quantitative evaluation (FID and IS). FID: lower is better; IS: higher is better.

Model	FID ↓	IS mean ↑	IS std
VAE	195.33	2.98	0.09
PercepVAE	174.83	2.93	0.08
cVAE-AdaGN	171.04	3.16	0.10
cVAE-Concat	163.23	3.54	0.10

4.4 Training Dynamics

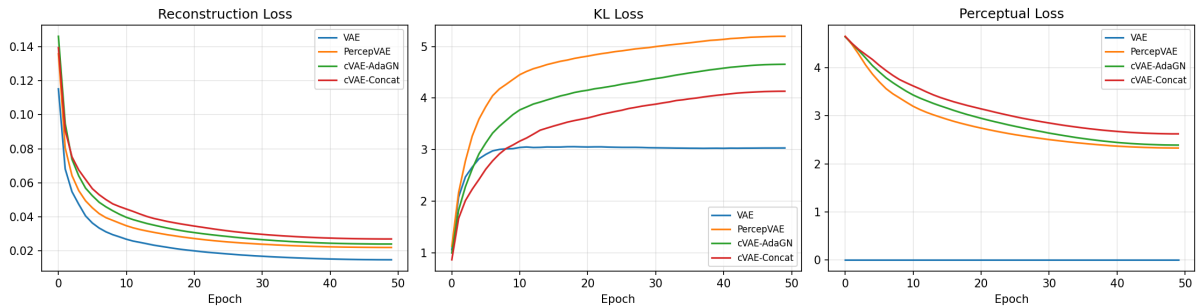


Figure 1: Experiment 1 — Training loss curves for all four VAE variants. The plot shows the evolution of the total loss, reconstruction term, and KL-divergence term over 50 epochs.

4.5 Qualitative Results

4.5.1 Random Samples

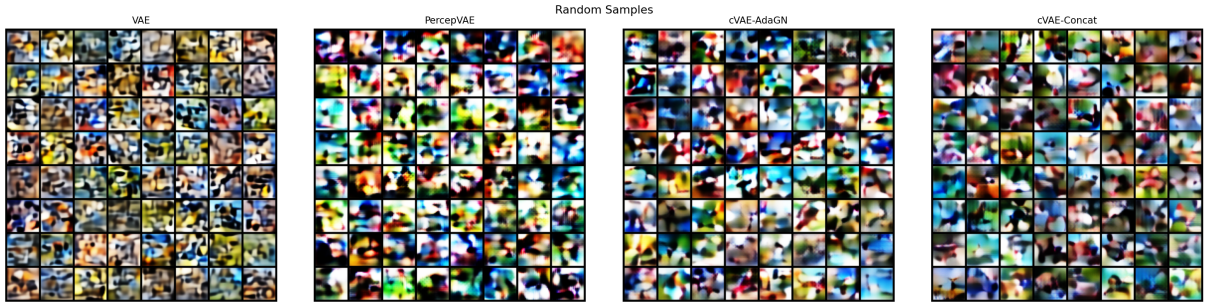


Figure 2: Experiment 1 — Randomly sampled images generated by decoding latent vectors $\mathbf{z} \sim \mathcal{N}(0, I)$.

4.5.2 Reconstructions

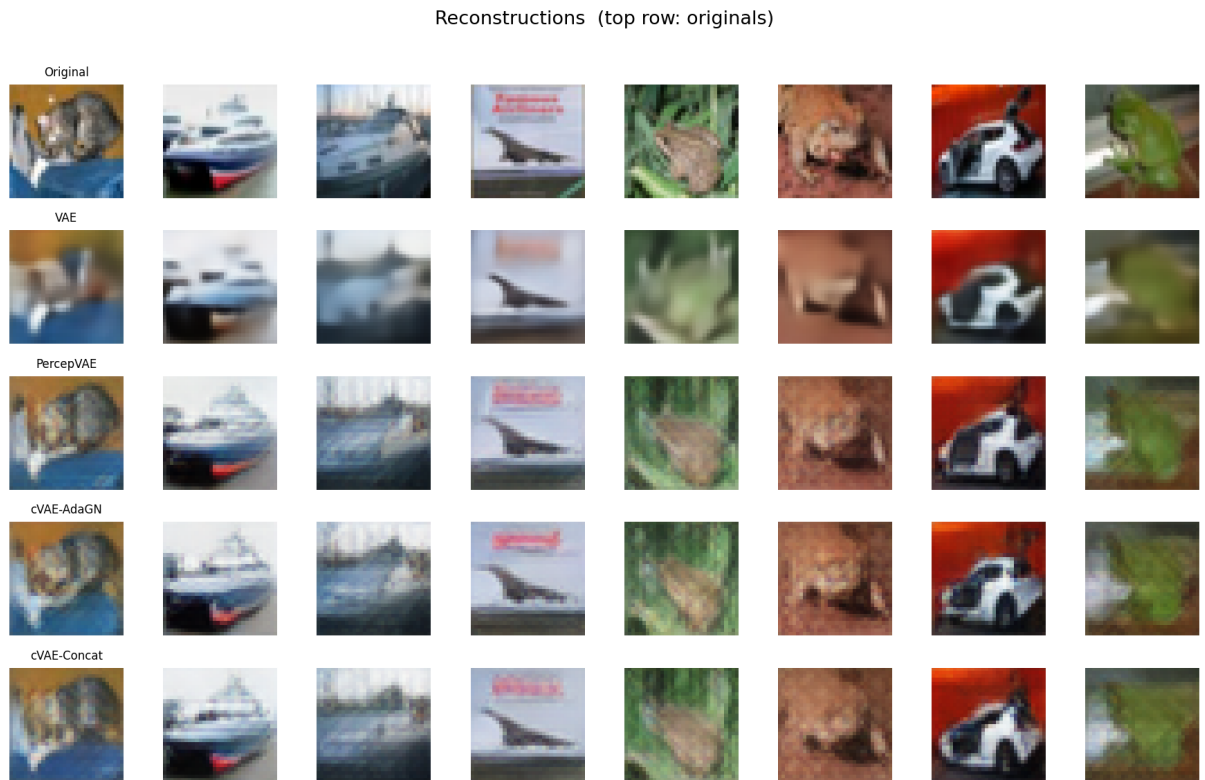


Figure 3: Experiment 1 — Input images (top row) and their reconstructions (bottom row). Reconstruction quality is noticeably better than free generation, reflecting the encoder's ability to infer informative posterior means.

4.5.3 Latent-Space Interpolations



Figure 4: Experiment 1 — linear interpolation between pairs of encoded images in latent space.

4.5.4 Latent-Space Visualisation

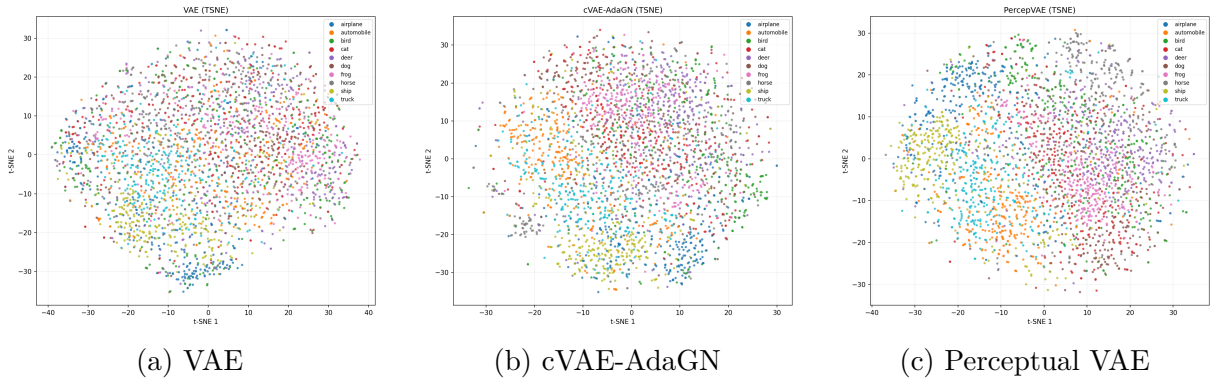


Figure 5: Experiment 1 — 2-D t-SNE projections of the latent space, coloured by class label. Tighter, better-separated clusters indicate that the model has learned class-discriminative structure in $\mathbf{z}$.

4.6 Analysis and Observations

- **Conditioning is decisive.** Both cVAE variants outperform the unconditional baselines by a substantial margin, with cVAE-Concat achieving the lowest FID (163.23) and the highest IS (3.54).
- **Perceptual loss provides moderate benefit.** PercepVAE reduces FID by ≈ 20 points vs. the vanilla VAE, but its IS slightly decreases, suggesting that perceptual regularisation may introduce a diversity-quality trade-off.
- **AdaGN vs. Concat.** cVAE-Concat outperforms cVAE-AdaGN on both FID and IS, indicating that direct feature concatenation is a more effective conditioning mechanism than normalisation-layer modulation at this scale.
- **Reconstruction vs. generation gap.** Reconstruction quality is visually good (see Figure 13), while free-sampling quality remains limited (Figure 12). This posterior-prior gap is a well-known VAE limitation.
- **Latent structure.** The t-SNE plots (Figure 15) show that conditional model form tighter, more separable clusters, consistent with improved IS scores.

5 Experiment 2

5.1 Motivation

The first experiment establishes the **baseline** performance of four VAE variants that differ in their conditioning and loss strategy, without self-attention. This provides a reference point for subsequent ablations.

5.2 Hyperparameters

Experiment 1 — Variable Hyperparameters

```
{
  "epochs"      : 50,
  "kl_weight"   : 1.0,    # Weight on KL-
                        divergence term
  "percep_weight" : 0.1,    # Weight on perceptual
                        loss
  "use_attention" : false,  # Self-attention
                        disabled
  "attn_heads"  : 1,
}
```

5.3 Evaluation Results

Table 2: Experiment 1 — Quantitative evaluation (FID and IS). FID: lower is better; IS: higher is better.

Model	FID ↓	IS mean ↑	IS std
VAE	326.79	1.69	0.03
PercepVAE	276.51	2.22	0.06
cVAE-AdaGN	394.40	2.03	0.03
cVAE-Concat	193.62	3.41	0.12

5.4 Training Dynamics

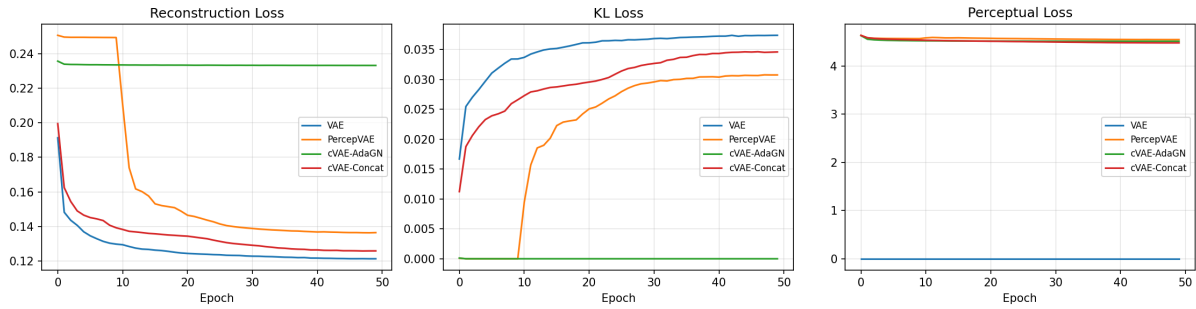


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5.5 Qualitative Results

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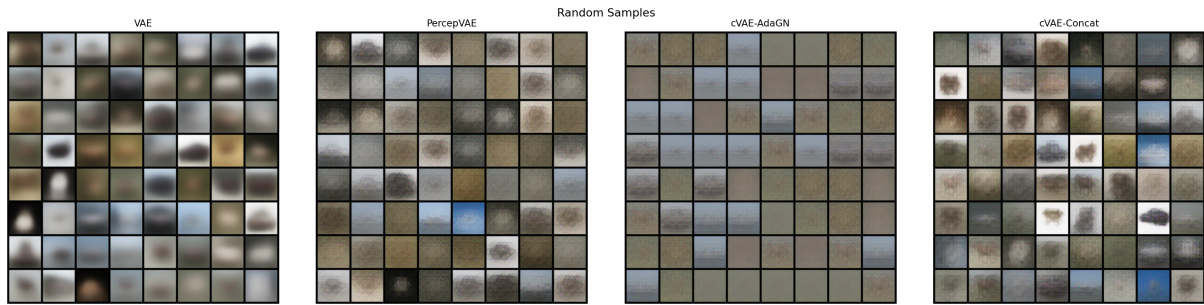


Figure 7: Experiment 1 — Randomly sampled images generated by decoding latent vectors $\mathbf{z} \sim \mathcal{N}(0, I)$.

5.5.2 Reconstructions

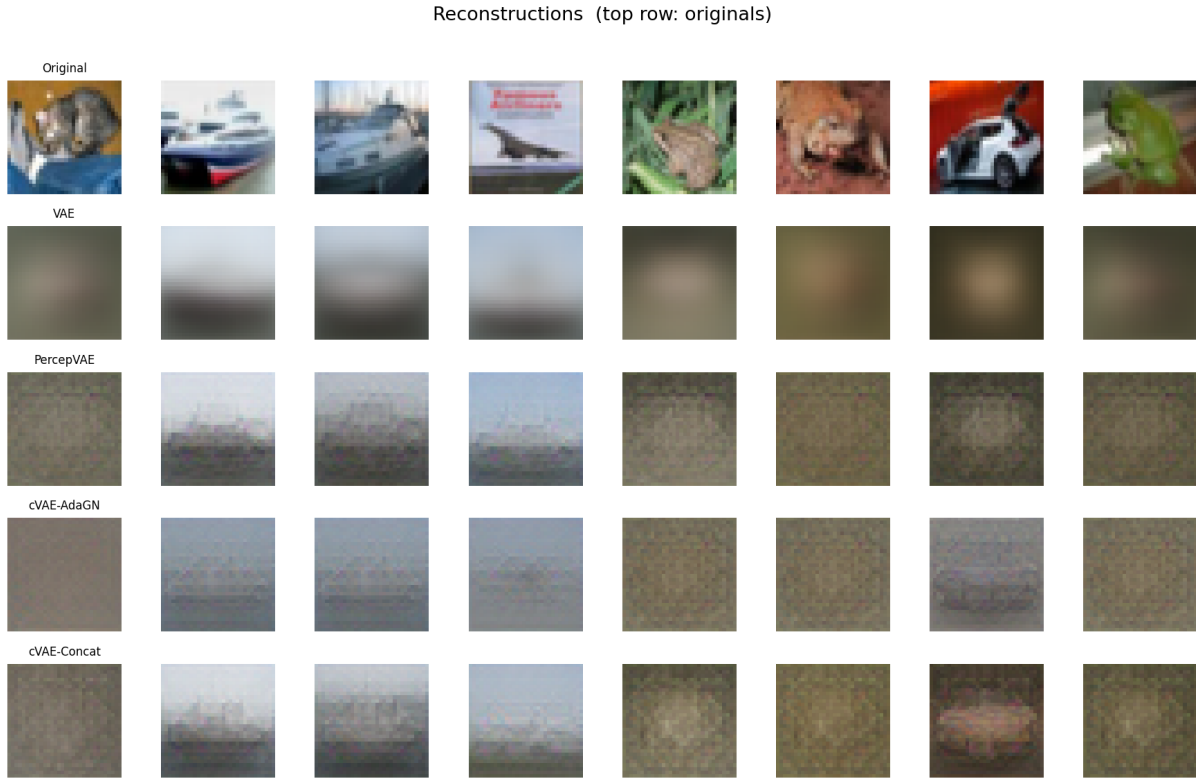


Figure 8: Experiment 1 — Input images (top row) and their reconstructions (bottom row). Reconstruction quality is noticeably better than free generation, reflecting the encoder’s ability to infer informative posterior means.

5.5.3 Latent-Space Interpolations

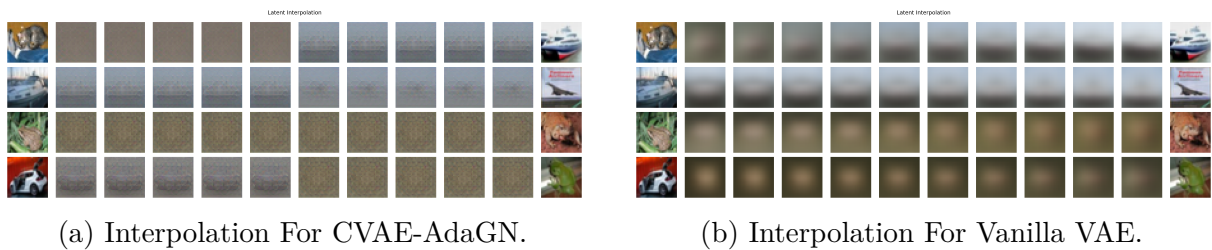


Figure 9: Experiment 1 — linear interpolation between pairs of encoded images in latent space.

5.5.4 Latent-Space Visualisation

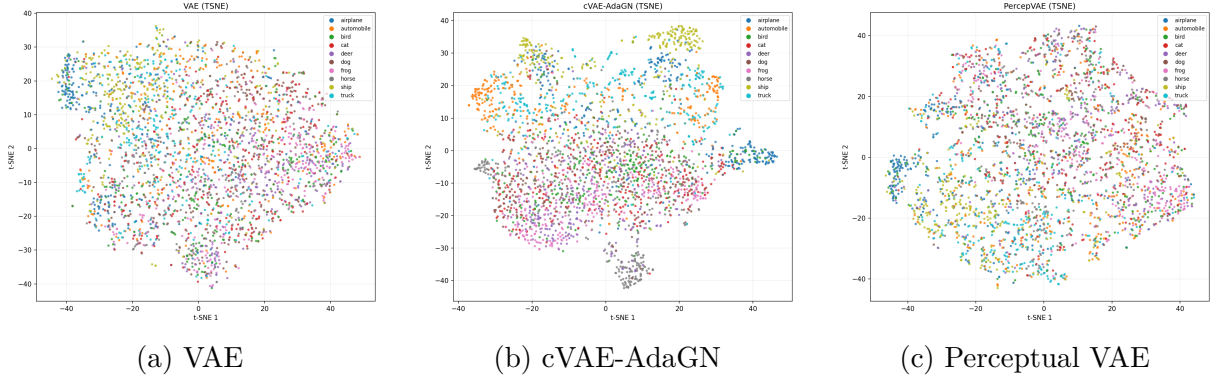


Figure 10: Experiment 1 — 2-D t-SNE projections of the latent space, coloured by class label. Tighter, better-separated clusters indicate that the model has learned class-discriminative structure in $\mathbf{z}$.

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6 Experiment 3

6.1 Motivation

The first experiment establishes the **baseline** performance of four VAE variants that differ in their conditioning and loss strategy, without self-attention. This provides a reference point for subsequent ablations.

6.2 Hyperparameters

Experiment 1 — Variable Hyperparameters

```
{
  "epochs"      : 30,
  "kl_weight"   : .1,    # Weight on KL-divergence
                        term
  "percep_weight" : 0.1,    # Weight on perceptual
                        loss
  "use_attention" : false,  # Self-attention
                        disabled
  "attn_heads"  : 1,
}
```

6.3 Evaluation Results

Table 3: Experiment 1 — Quantitative evaluation (FID and IS). FID: lower is better; IS: higher is better.

Model	FID ↓	IS mean ↑	IS std
VAE	326.79	1.69	0.03
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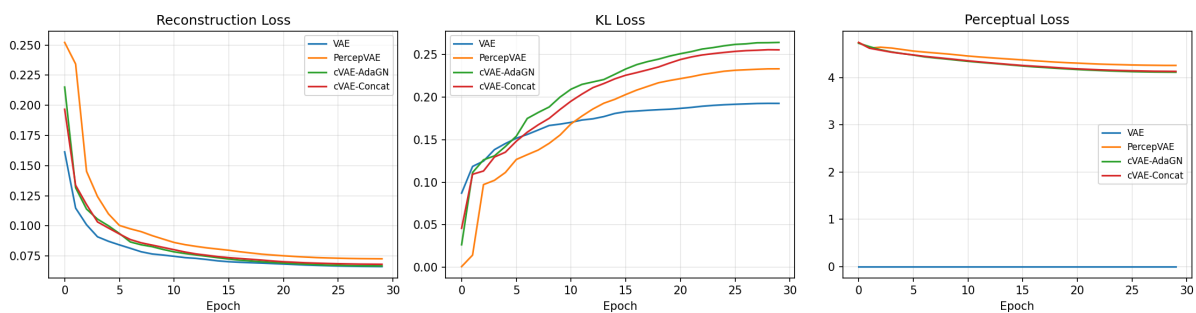


Figure 11: Experiment 1 — Training loss curves for all four VAE variants. The plot shows the evolution of the total loss, reconstruction term, and KL-divergence term over 50 epochs.

6.5 Qualitative Results

6.5.1 Random Samples

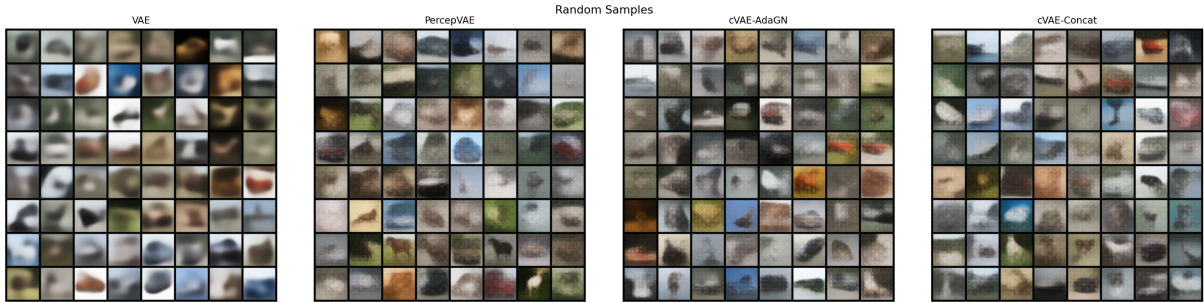


Figure 12: Experiment 1 — Randomly sampled images generated by decoding latent vectors $\mathbf{z} \sim \mathcal{N}(0, I)$.

6.5.2 Reconstructions

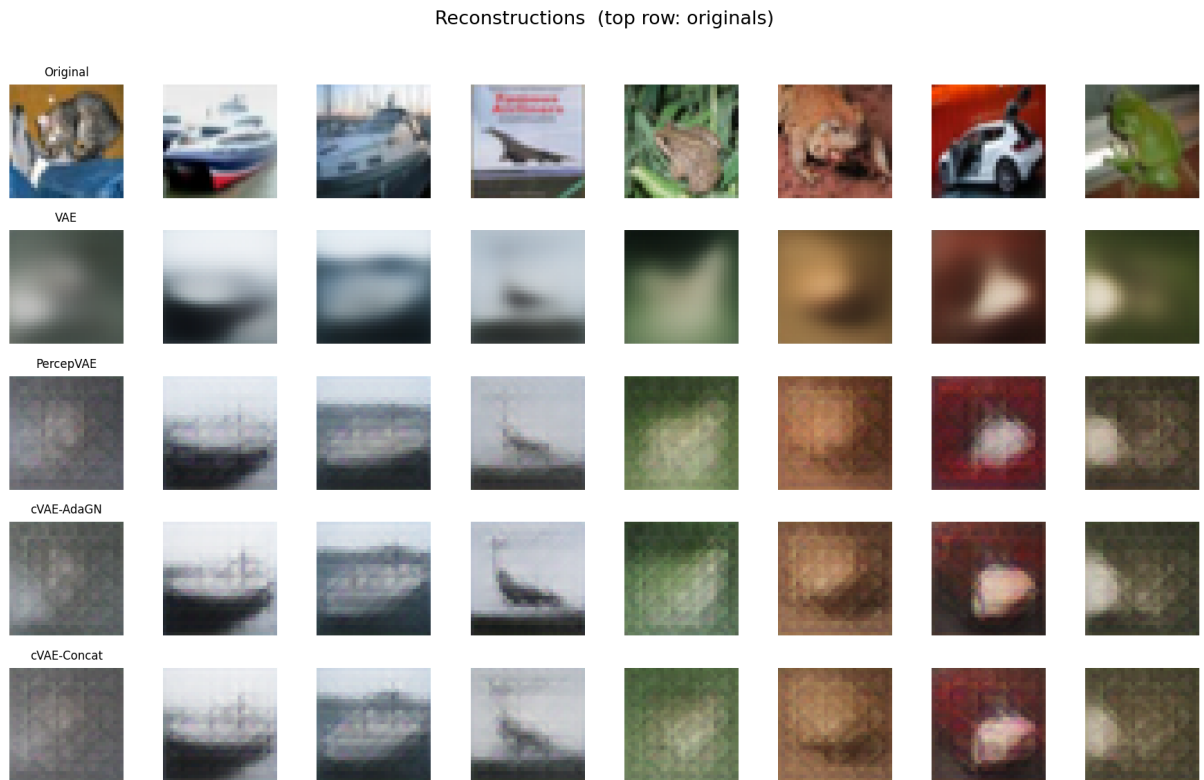


Figure 13: Experiment 1 — Input images (top row) and their reconstructions (bottom row). Reconstruction quality is noticeably better than free generation, reflecting the encoder’s ability to infer informative posterior means.

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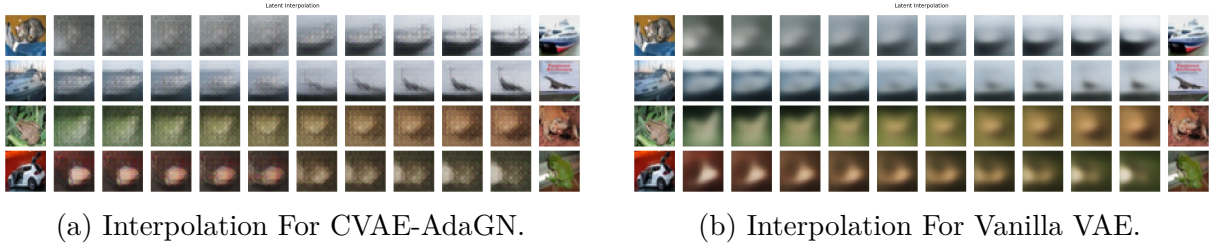


Figure 14: Experiment 1 — linear interpolation between pairs of encoded images in latent space.

6.5.4 Latent-Space Visualisation

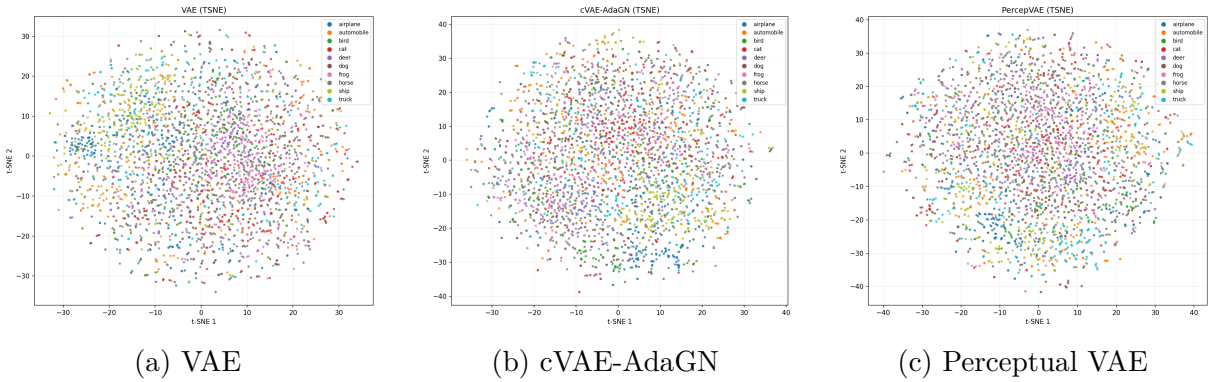


Figure 15: Experiment 1 — 2-D t-SNE projections of the latent space, coloured by class label. Tighter, better-separated clusters indicate that the model has learned class-discriminative structure in $\mathbf{z}$.

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- **Latent structure.** The t-SNE plots (Figure 15) show that conditional model form tighter, more separable clusters, consistent with improved IS scores.

7 Cross-Experiment Comparison

Content placeholder

Once all VAE experiments are complete, populate the summary table and figure below.

7.1 Summary Table — All VAE Experiments

Table 4: Summary of the best-performing VAE variant per experiment.

Exp.	Best Model	Key Change	FID ↓	IS ↑
1	cVAE-Concat	Conditioning strategy	163.23	3.54
2	—	—	—	—

7.2 FID Progression Across Experiments

Figure 16: FID of the best VAE variant per experiment, showing how successive design choices affect generation quality.

8 Conclusions

Content placeholder

This section will synthesise:

- The most impactful design choices for VAE generation quality
- The ceiling on generation quality achievable with the VAE objective alone
- Key structural properties of the learned latent space
- How these findings motivate the comparison with DDPMs (addressed in the companion report)

Based on Experiment 1 alone, several conclusions can already be drawn:

1. **Class conditioning is the single most impactful factor** evaluated so far, reducing FID by over 30 points relative to the unconditional VAE.
2. **Perceptual loss helps FID but not IS**, suggesting it encourages more realistic textures at the possible cost of diversity.
3. **The posterior-prior gap** remains the fundamental limitation of VAEs: the model reconstructs well but samples poorly. This is structurally different from DDPMs, where sampling and generation are the same operation.
4. **The latent space is interpretable and smooth**, as evidenced by coherent interpolations and class-separable t-SNE clusters. This is a property not easily available in DDPMs without additional inversion techniques.

A Experimental Protocol Checklist

Use this checklist when adding a new experiment to the report:

- ☐ State the motivation and hypothesis
- ☐ List all variable hyperparameters in a `configbox`
- ☐ List all model variants with a one-line description each
- ☐ Include the quantitative results table (FID, IS mean, IS std)
- ☐ Include the training loss figure
- ☐ Include the random-samples figure with observation
- ☐ Include the reconstruction figure with observation
- ☐ Include the latent-interpolation figure
- ☐ Include the latent t-SNE visualisation
- ☐ Write an analysis subsection with bullet-point observations
- ☐ Update the cross-experiment summary table (Section 7)

B Metric Definitions

Fréchet Inception Distance (FID). Let μ_r, Σ_r and μ_g, Σ_g be the mean and covariance of Inception-v3 pool3 features extracted from real and generated images respectively. Then:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right).$$

Inception Score (IS).

$$\text{IS} = \exp(\mathbb{E}_{\mathbf{x}} [D_{\text{KL}}(p(y|\mathbf{x}) \| p(y))]),$$

where $p(y|\mathbf{x})$ is the Inception classifier’s conditional label distribution and $p(y) = \int p(y|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}$.